Lemma. Given sequences $\{a_n\}$ and $\{b_n\}$, denote $A_n = \sum_{k=0}^{n} a_k$, $B_n = \sum_{k=0}^{n} a_k$.

Then, for any $q \in \mathbb{N}$,

$$\sum_{n=0}^{q} a_n b_n = \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q = \sum_{n=0}^{q-1} \left(\sum_{k=0}^{n} a_k \right) \cdot (b_n - b_{n+1}) + b_q \cdot \sum_{n=0}^{q} a_n .$$

In particular, if $0 \le p \le q$,

$$\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p$$

Proof. Since $a_n = A_n - A_{n-1}$, $(A_{-1} := 0)$

$$\sum_{n=0}^{q} a_n b_n = \sum_{n=0}^{q} (A_n - A_{n-1}) \cdot b_n = \sum_{n=0}^{q} A_n b_n - \sum_{n=0}^{q} A_{n-1} b_n$$

$$= \sum_{n=0}^{q} A_n (b_n - b_{n+1}) + \sum_{n=0}^{q} A_n b_{n+1} - \sum_{n=0}^{q} A_{n-1} b_n$$

$$= \sum_{n=0}^{q} A_n (b_n - b_{n+1}) + A_q b_{q+1}$$

$$= \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q .$$

Particular, $\sum_{n=p}^{q} a_n b_n = \sum_{n=0}^{q} a_n b_n - \sum_{n=0}^{p-1} a_n b_n$

$$= \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - \left(\sum_{n=0}^{p-2} A_n (b_n - b_{n+1}) + A_{p-1} b_{p-1} \right)$$

$$= \sum_{n=p-1}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_{p-1}$$

$$= \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_{p-1} b_{p-1} - A_{p-1} b_p + A_q b_q - A_{p-1} b_{p-1}$$

Cancle

Thm. If $\sum a_n$ Converges and $\{b_n\}$ is monotone bounded, then $\sum a_n b_n$ converges.

Proof. Since $\{b_n\}$ is monotone bounded, it is Convergent. put $b = \lim_{n\to\infty} b_n$, $A_n = \sum_{k=0}^{n} a_k$.
WLOG, Suppose $\{b_n\}$ is monotone decreasing. Then, $\{b_n - b\}$ Converges to Zero, and decreasing.
Let $M>0$ be a bounded of A_n. Given $\varepsilon>0$, put $N \in \mathbb{N}$ such that $n \geq N \Rightarrow |b_n - b| < \frac{\varepsilon}{2M}$.
By Abel's fomular, for all $q \geq p \geq N$,

$$\left| \sum_{n=p}^{q} a_n(b_n - b) \right| = \left| \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q(b_q - b) - A_{p-1}(b_p - b) \right|$$

$$= M \cdot \left[\sum_{n=p}^{q-1} |\underbrace{b_n - b_{n+1}}_{\geq 0}| + |\underbrace{b_q - b}_{\geq 0}| + |\underbrace{b_p - b}_{\geq 0}| \right]$$

$$= M \cdot \left[\sum_{n=p}^{q-1} (b_n - b_{n+1}) + (b_q - b) + (b_p - b) \right]$$

$$= M \cdot \left[b_p - b_q + b_q - b + b_p - p \right]$$

$$= 2M(b_p - b) < 2M \cdot \frac{\varepsilon}{M} = \varepsilon,$$

Therefore, by the Cauchy Criterion for Series, $\sum_{n=0}^{\infty} a_n(b_n - b)$ Converges.
Moreover, Since $\sum_{n=0}^{\infty} a_n$ Converges, So is $b \cdot \sum_{n=0}^{\infty} a_n$. Finally,

$$\sum_{n=0}^{\infty} a_n(b_n - b) + \sum_{n=0}^{\infty} a_n \cdot b = \sum_{n=0}^{\infty} a_n b_n$$

Converges, as sum of two Convergent series. ∎

Note: If $b=0$, then it is enough to converges that $\sum a_n$ bounded, not necessary converges.